Chapter 8

OSPF

OSPF (Open Shortest Path First) is the de-facto open standard routing protocol for large networks. Table I shows the basic characteristics of OSPF.

TABLE I Routing characteristics of OSPF

Metric	Cost
Interior or Exterior?	Interior
Distance vector or Link-state?	Link-state
Singlepath or Multipath?	Multipath
Broadcast or Multicast?	Multicast
Flat or Hierarchical?	Hierarchical
Classful or Classless?	Classless

Advantages of OSPF

- Widely supported open standard—runs on most platforms
- Classless protocol—full support for classless addressing and therefore classless routing
- Scalable protocol—can be used on very large networks
- Extensible—periodically updated via RFCs to keep pace with evolution of the Internet
- Highly configurable to manage bandwidth utilized by routing protocol traffic
- No hop count limitation
- Fast convergence with small routing updates
- Not prone to routing loops
- Routing traffic sent via multicasts
- Optional authentication of routing packets—prevents rogue routers from advertising unauthorized routes

Disadvantages of OSPF

- Perceived as relatively complex to configure (not necessarily true)
- Runs over IP only
- Does not support unequal cost multipath routing (can be mitigated through metric manipulation)
- Summarizes routes at area borders only
- May require renumbering of network in order to obtain desired summarization of routes

The differences between OSPF and EIGRP are given in Table II.

TABLE II Differences between OSPF and EIGRP

	EIGRP	OSPF
1	It supports maximum 255 routers in the network. However, the default is 100 routers. (highly scalable)	Supports unlimited number of routers
2	Fast convergence due to feasible successor	Fastest convergence speed due to the area concept
3	Cisco proprietary protocol and can be implemented only in Cisco routers.	Open standard protocol and can be implement in any router.
4	It calculates the metric In terms of bandwidth and delay (default).	It calculates the metric In terms of bandwidth only.
5	EIGRP works on DUAL(Diffusing Update Algorithm) Algorithm.	EIGRP works on Dijkstra Algorithm.
6	It maintains the best route and some other alternative routes for each destination.	It maintains the best route in routing table and all routes in database table.
7	It is basically use for medium to lager size organization in the network [1].	It is basically use for lager size organization in the network [1].
8	Administrative distance 90	Administrative distance 110
9	Easy to implement	The implementation is complicated

8.1 OSPF Terminology

Area

An area in OSPF is a logical grouping of routers that share the exact same link-state database. Areas are one of several mechanisms for controlling the resources consumed by routing updates. It is not necessary for all routers in an autonomous system to receive detailed link-state information from all other routers in the system, and areas provide a means to limit such traffic. OSPF allows for several types of areas, depending on the network configuration.

Backbone area

This is the only mandatory area in an OSPF network. The backbone area is always designated area 0 (or 0.0.0.0.). There can be one, and only one area 0. Normally, all other areas connect to area 0 and pass routing traffic bound for other areas through area 0. Link-state advertisements (will be discussed later) are summarized before being advertised to other areas.

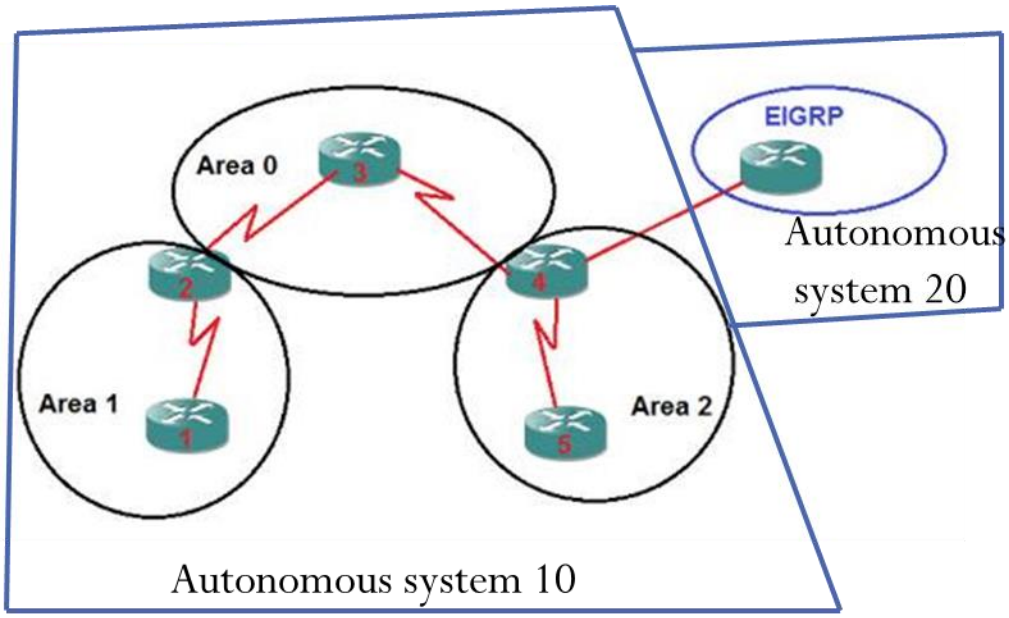

Fig. 8.1 Autonomous system and Area

In Fig. 8.1, the relationship between autonomous system and area is shown. There are two autonomous systems with numbers 10 and 20. The autonomous system 10 has been divided into three areas: 0, 1 and 2. On the other hand, the autonomous system 20 has only one area.

Router Types

Router types go hand-in-hand with area types and are matched to the role they play within the area.

Internal Router

An internal router has all interfaces contained within a single area and does not connect to any other areas. In Fig. 8.1, the routers 1 and 5 belong to this group. All internal routers of an area have the identical link-state database.

Area Border Router (ABR)

An ABR connects at least two areas. It maintains a separate link-state database for each area it connects to. Link-state summarization and route summarization occur at border routers. The routers 2 and 4 belong to this group.

Backbone Router

This is a router that has at least one interface in area 0. A backbone router that also has an interface in another area is also considered an ABR. Routers 2, 3 and 4 are backbone router.

Autonomous System Boundary Router (ASBR)

An ASBR has at least one interface connecting to another autonomous system (or redistributes a static route). Since this autonomous network can be non-OSPF, the ASBRs are capable of route redistribution, which means that these routers can import routing information from non-OSPF networks and distribute it in the same OSPF network for which it is responsible, as well as taking

routing information from the OSPF network and distributing it into the non-OSPF system. Router 4 is an ASBR.

All OSPF networks have at least 1 area. The default area is given the number 0 and functions as the backbone as additional areas are added. All other areas connect through area 0, and each defined area has its own link-state database.

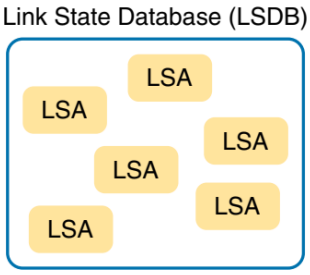

Fig. 8.2 LSDB and LSAs

OSPF data structure and packets

Link state Advertisement (LSA)

A data structure with some specific information about the networks. It is used to describe information about a router, or a network or a subnet. Details will be discussed later.

Link state database (LSDB)

It is a collection of all LSAs known to a router (see Fig. 8.2). Every router maintains its own link-state database. When the network is converged, all routers in an area have the exact same link-state database.

Hello

Hello packets are used to establish and maintain adjacency with other OSPF routers.

Database description (DBD)

This packet contains an abbreviated list of the sending router's link-state database and is used by receiving routers to check against the local link-state database.

Link state request (LSR)

Receiving routers can then request more information about any entry in the DBD by sending a link-state request (LSR).

Link-state update (LSU)

LSU packets are used to reply to LSRs and to announce new information. LSUs contain seven different types of link-state advertisements (LSA).

Link-state acknowledgement (LSAck)

When an LSU is received, the router sends a link-state acknowledgment (LSAck) to confirm receipt of the LSU

Conditions for neighborship

During neighbor discovery, a router sends a Hello packet through all its interfaces. It contains all information that two directly connected routers must match to be neighbor of each other. Following parameters need to be identical for two routers to become neighbors

Network mask—net mask of the sending router

Subnet number—derived using the subnet mask and each router's interface Internet Protocol (IP) address

Area ID—area ID of the sending interface

Hello interval—how often Hello packets are transmitted

Dead interval—how long to wait for Hello packets before terminating neighbor

Authentication type and password—optional

Stub area flag—specifies the type of stub area, if applicable

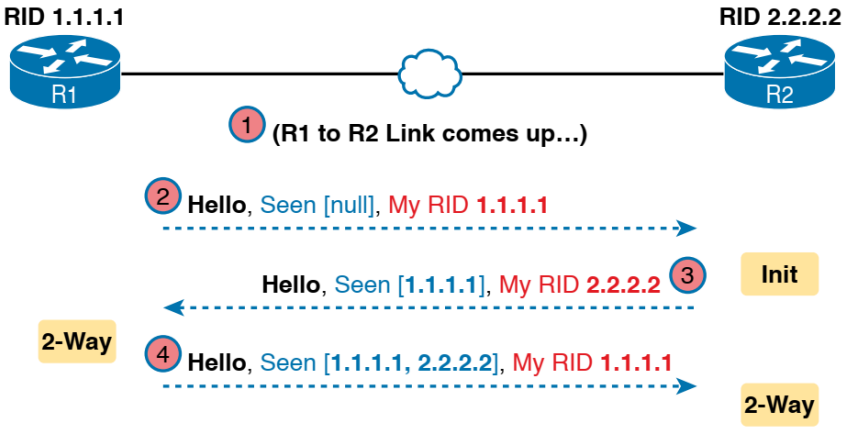

Fig. 8.3 Steps of OSPF neighbor discovery

8.2 OSPF Neighbor Discovery

The scenario begins with the link down, so the routers have no knowledge of each other as OSPF neighbors. The steps of the neighbor discovery are shown in Fig. 8.3 and described below:

1. Link between R1 and R2 comes up
2. R1 sends the first Hello to multicast IP address 224. 0.0.5, so R2 learns of the existence of R1 as an OSPF router. At that point, R2 lists R1 as a neighbor, with an interim beginning state of init.
3. R2 sends back a Hello which tells R1 that R2 exists, and it allows R1 to move through the init state and quickly to a 2-way state.

4. R2 receives the next Hello from R1, and R2 can also move to a 2-way state

8.3 RID, DR and BDR

Router ID (RID)

Each Router in an OSPF network needs a unique OSPF Router ID. The OSPF Router ID is used to provide a unique identity to the OSPF Router. OSPF Router ID is an IPv4 address (32-bit binary number) assigned to each router running the OSPF protocol. OSPF Router ID should not be changed after the OSPF process has been started and the OSPF neighborships are established.

OSPF Router ID selection algorithm works as below (see Fig. 8.4).

- Any manually configured OSPF Router ID in OSPF Process is selected as the OSPF Router ID.
- If there is no configured OSPF Router ID, the highest IP address on any of the Routers Loopback Interfaces is selected as the OSPF Router ID.
- If there is no Loopback Interfaces configured, the highest IP address on its active interfaces is selected as the OSPF Router ID.

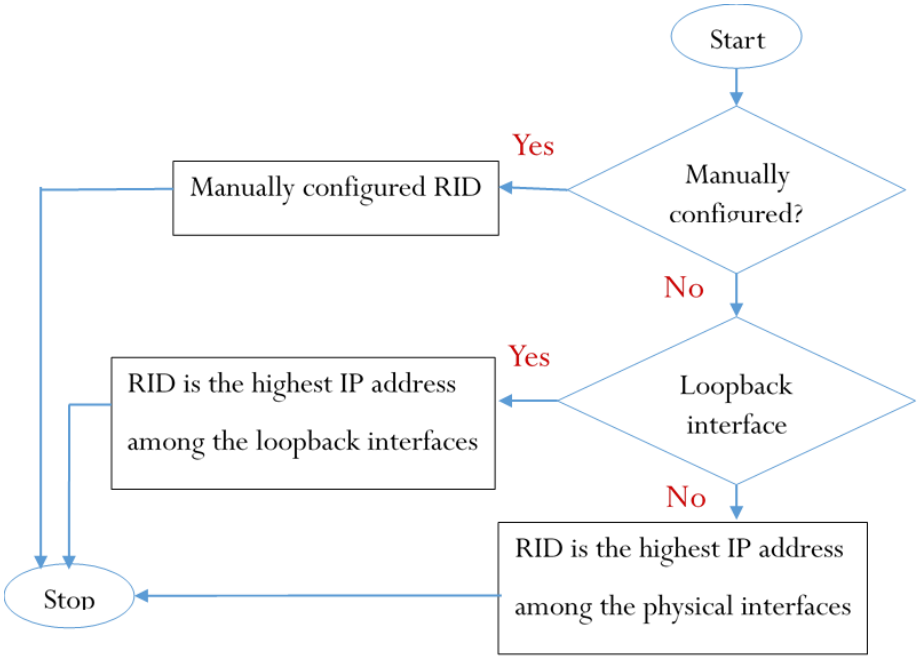

Fig. 8.4 Flow chart of router ID selection

DR and BDR

Before describing DR and BDR, let's first describe two types of networks: point-to-point and broadcast networks.

A network which connects two routers directly to each other is called a **point-to-point network.** In Fig. 8.5, R1 is directly connected to R2, R3, and R4. Thus, the networks 10.1.12.0/24 (R1-R2), 10.1.13.0/24 (R1-R3) and 10.1.14.0/24 (R1-R4) are all point-to-point networks.

A network which connects more than two routers is called a **broadcast network**. Such connection among more than two routers are done with the help of a layer 2 devices such as switch. In Fig. 8.6, four routers are connected the network 10.1.1.0/24, hence this is a broadcast network.

In broadcast network, the routers are divided into three categories: designated router (DR), backup designated router (BDR) and DROthers. Any two neighbor in such network do not exchange update directly to each other. Instead all such exchange is done via a router named DR for the sake of reducing traffic. However, if the DR fails, BDR takes over as a DR and a new BDR is elected. Each router in a broadcast network is configured with a priority which is an integer between 1 to 65535. The default priority of all routers is 1. Among all the routers of a broadcast network, the router with the highest priority is elected as the DR, while the second highest is elected as BDR. However, if all routers are having the same priority, the router having the highest RID is elected as the DR, while the router with the second highest RID is elected as the BDR. The rest of the routers in the network are call DROthers. In Fig. 8.6, Since R4 and R3 have the highest and second highest RID, they are elected as the DR and BDR, respectively, under the consideration that all routers have the same priority.

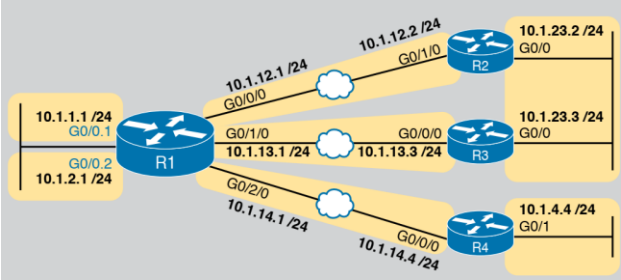

Fig. 8.5 Point-to-point network

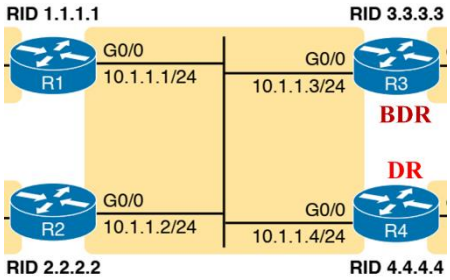

Fig.8.6 Broadcast network

Each DROthers router will exchange routing information only with the DR and BDR, instead of exchanging updates with every router on the network segment. DR will then distribute topology information to every other routers inside the same area, which greatly reduces OSPF traffic. To send routing information to a DR or BDR the multicast address of 224.0.0.6 is used. DR sends

routing updates to the multicast address of 224.0.0.5. If DR fails, BDR takes over its role of redistributing routing information. Every router on a network segment will establish a full neighbor relationship with the DR and BDR. Non-DR and non-BDR routers will establish a two way neighbor relationship between themselves.

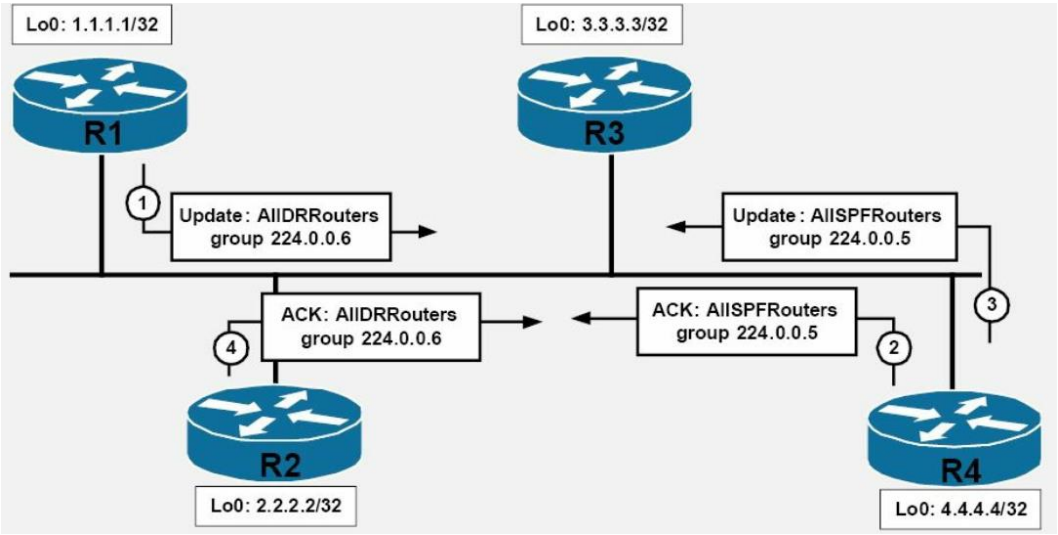

Fig. 8.7 Illustration of update exchange through DR and BDR

It is important to understand the sequence of message exchanges when a DR and/or a BDR router have been elected. As an example, imagine a Broadcast network with four routers, which are R1, R2, R3, and R4 (see Fig. 8.7). Assume that R4 has been elected as DR, and R3 has been elected as BDR. R2 and R1 are neither DR nor BDR and are therefore referred to as DROther routers in Cisco OSPF terminology. A configuration change is made on R1, and R1 then sends an update to the AllDRRouters (DR and BDR) Multicast group address 224.0.0.6. R4, the DR receives this update and sends an acknowledgement back to the AllSPFRouters Multicast group address 224.0.0.5. R4 then sends this update to all other DROthers routers using the AllSPFRouters Multicast group address. This update is received by the other DROthers router, R2, and R2 sends an acknowledgement to the AllDRRouters Multicast group 224.0.0.6.

8.4 Wildcard Mask

Wildcard masks are used to specify a range of network addresses. They are commonly used with routing protocols such as OSPF and access lists.

Just like a subnet mask, a wildcard mask is 32 bits long. It acts as an inverted subnet masks, but with wildcard mask, the 0 bits indicate that the corresponding bit position must match in all the network IP address in the range. The 1 bit indicates that the corresponding bit position does not have to match the bit position in the IP addresses of all networks.

Suppose that we want to create a range of IP addresses which consists of only one address 192.168.3.0. Since it consists of a range where there can be only one address, so all bits (32 bits)

of all the IP addresses in the range must match. Since there can not have two IP address having all the bits same, there will be only one IP address in the range. Since all bits must match, so all the 32 bits must be 0. Thus, the wildcard mask is 00000000. 00000000. 00000000. 00000000. The wildcard mask in decimal is 0.0.0.0.

Suppose that we want to create a range which all IP addresses of the 192.168.3.0/24 network will belong to. Since the prefix is 24, the first 24 bits are the network bits while the rest 8 bits are host bits. Since all hosts of a network must have the same network bits, all IP addresses of the range must have the same first 24 bits. So, make the first 24 bits 0 and the rest 8 bits 1. That is, the wildcard is 00000000.00000000.00000000.11111111 =0.0.0.255. It can also be achieved by subtracting the subnet mask of the network from 255.255.255.255. The subnet mask of the given network is 255.255.255.0 as the prefix length is 24. Subtracting the subnet mask from 255.255.255.255, we get the corresponding wildcard mask 0.0.0.255.

Now consider a range of IP addresses from 192.168.3.4 to 192.168.3.13

To find the wildcard mask to create the range, represent the first and the last IP addresses of the range in binary.

11000000.10101000.00000011.00000100

11000000.10101000.00000011.00001101

We see that the 29th bit is the first dissimilar bit. So make the first 28 (32-4) bits 0 and the last 4 bits 1. The wildcard mask becomes 00000000. 00000000. 00000000.00001111 = 0.0.0.15

Actually, the wildcard mask creates a range from 192.168.3.0 to 192.168.3.15.

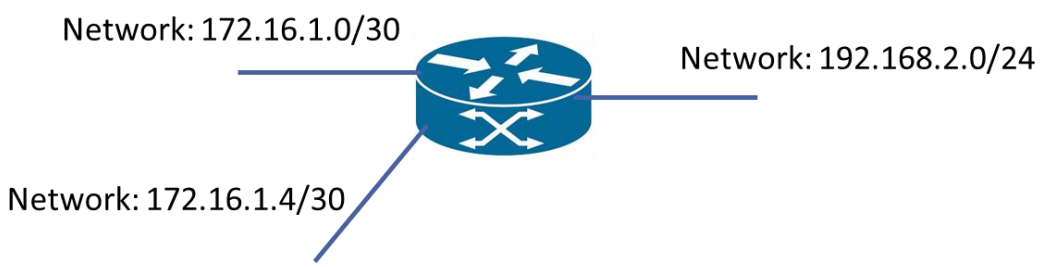

Fig. 8.8 OSPF configuration

8.5 OSPF Configuration

The command for configuring OSPF in a router, we have to use the following command

```
R1(config)# router ospf process_id
R1(config-router)#network network_IP  WCM  area area_id
```

,where the `process_id` is any integer in the range 1 to 65535. All routers of an area do not need to have the same `process_id`. The *`network_IP`* is the IP address of the network which is directly connected to any interface of the router. WCM is the wildcard mask. The `area_id` is any integer in the range 0-4294967295.

Let's consider an example of a router shown in Fig. 8.8. The router is connected to three networks 172.16.1.0/30, 172.16.1.4/0 and 192.168.2.0/24. One of several ways of configuring OSPF in the router is given by

```
R1(config)# router ospf 10
R1(config)# router-id  1.1.1.1
R1(config-router)#network 192.168.2.0    0.0.0.255  area 0
R1(config-router)#network 172.16.1.0   0.0.0.3  area 0
R1(config-router)#network 172.16.1.4    0.0.0.3  area 0
```

The first line configures OSPF in the router with process ID 10.

The second line sets RID to 1.1.1.1.

The fourth line configures OSPF for the network 192.168.2.0 whose wildcard mask 0.0.0.255. It also mentions that the interface to which this network is connected belongs to area 0. To get the wildcard mask of the network 192.168.2.0/24, we can just subtract the subnet mask of the IP, 255.255.255.0, from 255.255.255.255. It gives 0.0.0.255. Similarly, for the network 172.16.1.0/30, the subnet mask is 255.255.255.252, we can get its wildcard mask as

255.255.255.255

255.255.255.252

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0. 0. 0 .3

Exercise

Configure the following network using OSPF. Also verify the configuration using the verification commands.

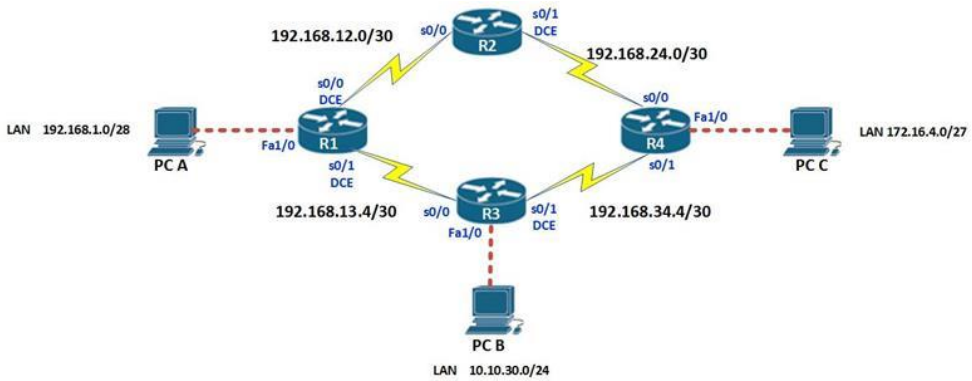

Fig. 8.9 Exercise I